Polluture

Ecological Impact Mapper · Kolkata

Compare | Clear All | Reset

ZONES 1

Draw New Zone

Zone 1 5.85 ha

MAP LAYER

Street | Satellite | Hybrid

ZONE 1

Land Type

Wetland / Lake | Dense Vegetation | Agricultural | Open / Bare Land

Floors 5

Residents 51

Horizon 25 yrs

Zone area: 5.85 ha

Calculate Impact

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ZONES 1

Draw New Zone

Zone 1 — 5.85 ha ✓

MAP LAYER

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ZONE 1

Land Type

Wetland / Lake | Dense Vegetation | Agricultural | Open / Bare Land

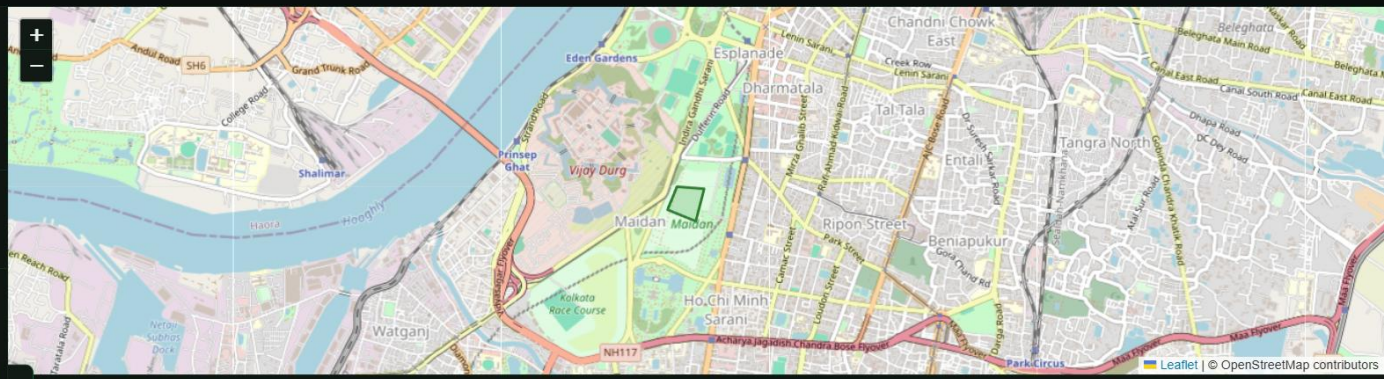
Floors: 5

Residents: 51

Horizon: 25 yrs

Zone area: 5.85 ha

Calculate Impact



Impact Report

Zone 1 · vegetation — 5.85 ha

ECOLOGICAL DEBT

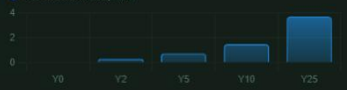
₹ 3,37,430

300	3.71m	+5.67°C
AQI Y25	GW drop	Heat rise

PROJECTIONS OVER 25 YEARS

AQI over time

Groundwater drop (m)



Temperature rise (°C)

25-YEAR NARRATIVE

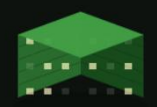
Year 0
Year 0: Canopy cover removed. Urban surface albedo drops — temperatures begin rising. 878 tonnes of carbon sequestration capacity is lost forever.

Year 4
Year 4: PM2.5 levels climb as natural air filtration disappears. Nearest AQI station registers 200 — in the 'Poor' category.

Year 9
Year 9: Local rainfall patterns shift due to loss of evapotranspiration. Droughts become more frequent. 468 mm/year less local moisture.

Year 15
Year 15: Soil erosion accelerates without root systems. The riverine sediment load increases by 117 tonnes/year.

Year 25
Year 25: 51 families live in an urbanized heat trap. Cooling costs surge. The area's ecological credit is fully exhausted — nature cannot self-recover.



5F

Polluture

Ecological Impact Mapper · Kolkata

Compare | Clear All | Reset

ZONES 3

Draw New Zone

- Zone 1 — 5.85 ha
- Zone 2 — 11.22 ha
- Zone 6 — 30.36 ha

MAP LAYER

Street | Satellite | Hybrid

ZONE 2

Land Type

Wetland / Lake | Dense Vegetation | Agricultural | Open / Bare Land

Floors: 5

Residents: 38

Horizon: 12 yrs

Zone area: 11.22 ha

Calculate Impact

Impact Report

COMBINED ECOLOGICAL DEBT ₹ 21,04,151

Zone 6 · wetland — 30.36 ha

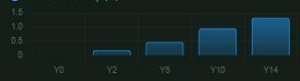
ECOLOGICAL DEBT ₹ 15,80,446

AQI Y14	GW drop	Heat rise
352	1.35m	+4.32°C

PROJECTIONS OVER 14 YEARS

AQI over time

Groundwater drop (m)



Temperature rise (°C)

14-YEAR NARRATIVE

Year 0
Year 0: The wetland is cleared for construction. Monsoon runoff now has nowhere to drain — seasonal flooding risk increases by 40%.

Year 2
Year 2: Groundwater recharge has dropped sharply. The local water table falls, and 27 residents report dry taps in summer.

Year 5
Year 5: AQI regularly crosses unsafe thresholds. Children in the area report elevated respiratory illness rates. The concrete heat island adds 0.8°C locally.

Year 8
Year 8: Biodiversity loss is now permanent — 364 migratory bird species no longer use this corridor. Urban runoff causes downstream eutrophication.

Year 14
Year 14: The ecological debt compounds. Remediation now would cost multiples of the original development. 89 residents live in degraded urban heat island conditions year-round.

Zone 2 · bare — 11.22 ha

ECOLOGICAL DEBT ₹ 1,86,275

AQI Y12	GW drop	Heat rise
192	0.17m	+1.41°C

PROJECTIONS OVER 12 YEARS

AQI over time

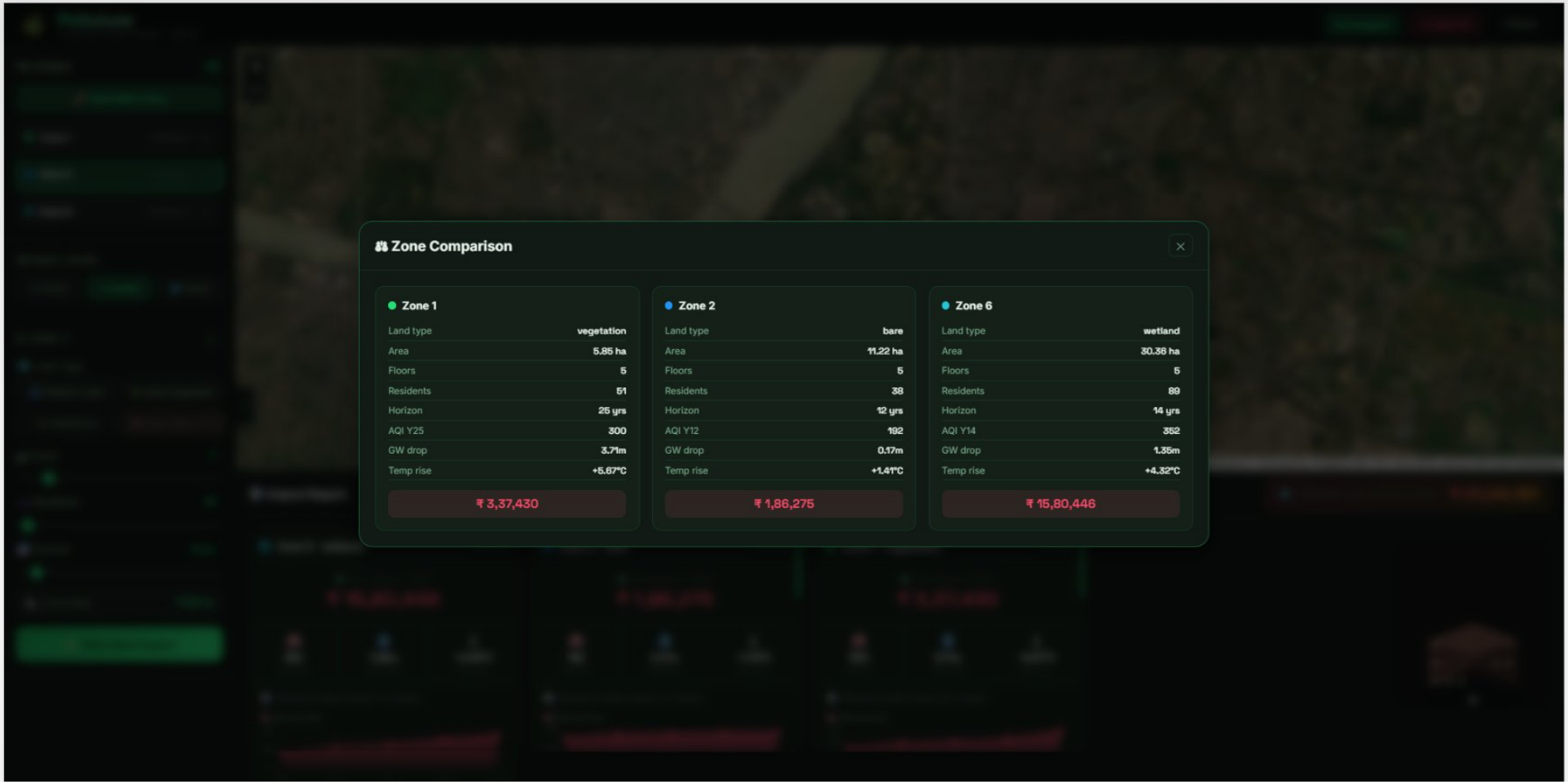
Zone 1 · vegetation — 5.85 ha

ECOLOGICAL DEBT ₹ 3,37,430

AQI Y25	GW drop	Heat rise
300	3.71m	+5.67°C

PROJECTIONS OVER 25 YEARS

AQI over time


Zone Comparison
Zone 1
Land type vegetation
Area 5.85 ha
Floors 5
Residents 51
Horizon 25 yrs
AQI Y25 300
GW drop 3.71m
Temp rise +5.67°C
₹ 3,37,430
Zone 2
Land type bare
Area 11.22 ha
Floors 5
Residents 38
Horizon 12 yrs
AQI Y12 192
GW drop 0.17m
Temp rise +1.41°C
₹ 1,86,275
Zone 6
Land type wetland
Area 30.36 ha
Floors 5
Residents 89
Horizon 14 yrs
AQI Y14 352
GW drop 1.35m
Temp rise +4.32°C
₹ 15,80,446